

Mahmoud Adel Abdulwahab

Software Engineer (Full Stack)

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 3 years of experience designing and building server-side systems in Node.js and TypeScript (NestJS, Express). Experienced in distributed microservices, event-driven messaging, REST API design, authentication and role-based access control, workflow orchestration, and integrating LLMs into production services, with close attention to reliability and data consistency. Works across MongoDB, PostgreSQL, and SQL Server, with full stack delivery in Angular, React, and Next.js. Has delivered enterprise platforms and client systems for companies across the Gulf and Egypt. Native Arabic speaker with professional English.

PROFESSIONAL EXPERIENCE

Software Engineer

Aug 2025 – Present

Contellect Technologies Inc. | Cairo, Egypt

- Develop and maintain a 50+ repository enterprise content management (ECM) platform of distributed Node.js/TypeScript microservices (NATS, ActiveMQ, KrakenD API gateway) with an Angular front end, designing and documenting REST APIs with OpenAPI for content, workflow, audit, and file-management modules.
- Re-architected the platform's AI chat assistant (Vercel AI SDK; Azure OpenAI, OpenAI, Anthropic Claude) from a single tool-loop agent capped by the 128-tool limit to a parent agent dispatching to domain-scoped sub-agents.
- Built AI usage auditing that lets administrators track tokens, messages, and responses across all users, on top of the user-activity and audit-log subsystem (MongoDB, Elasticsearch) used for compliance and incident investigation.
- Implemented Keycloak (OAuth 2.0 / OpenID Connect) authentication and role-based access control across services, consolidating per-service permission checks into a single policy layer.
- Delivered legal hold for a Records Management business app: held records become read-only across services (no edit, delete, move, or new version), controlled by a dedicated permission.
- Automated business processes with Camunda Zeebe (BPMN) and Temporal, including document approval and routing and an HR Onboarding & Offboarding app.
- Made a message consumer idempotent to eliminate duplicate records under concurrent publishes, and fixed chunked-upload pause/resume so large files continue from the last uploaded chunk instead of restarting from 0%.
- Built Angular features (dynamic Form.io forms, dashboards, workflow interfaces) released behind Unleash feature flags, covered by unit and integration tests in Jest and Supertest.
- Built an AI-assisted engineering workflow with Claude Code and MCP integrations (Jira, Bitbucket) that takes tickets through investigation, implementation, testing, and review to a pull request.

Full Stack Developer

Feb 2024 – Jul 2025

Slash Solutions | Remote

- Built backend services and REST APIs with Node.js and NestJS for booking, ERP, and internal business systems delivered to clients in Egypt and the Gulf.
- Integrated payment gateways (Moyasar, Tap, Stripe, PayPal, Paymob) into client checkout and billing flows.
- Implemented JWT authentication and role-based access control for multi-role business users.
- Integrated third-party SMS, email, and maps APIs, and built reporting with PDF and Excel exports.
- Designed data models across PostgreSQL, MySQL, SQL Server, and MongoDB, and built React and Angular admin interfaces for the backend services.

PROJECTS

Saknly: Arabic-First Property Marketplace

[Live](#) · [Code](#)

- Built an Arabic-first property marketplace for sale, rent, and student housing with Node.js/Express, Next.js 15 (App Router, TypeScript), MUI with full RTL, and MongoDB Atlas, including Google sign-in (Passport), email confirmation, and token revocation on logout.
- Built listing publishing with saved drafts and photo processing (Multer, sharp, Cloudinary), an admin moderation queue with bulk approval, and cascading cleanup of photos, comments, and inquiries when a listing is deleted.
- Implemented shareable URL-driven search (location, price, area, rooms, amenities) with pagination and a Leaflet map view, plus a Google Gemini chatbot that answers questions about real listings and prices in Arabic; API tested with Vitest and Supertest.

3M Mile: Car Care & Detailing Platform

app.3mmile.io

- Built and launched an Arabic (RTL) website with an admin dashboard and CMS for a car-care business in Jeddah using Next.js, MongoDB/Mongoose, and Cloudinary.
- Built the CMS so the business manages service pages (paint protection film, nano ceramic, heat insulation, wraps) and media through the admin dashboard.

Pionner: Bilingual E-Commerce Store for Saudi Arabia

[Live](#) · [Code](#)

- Built a bilingual (Arabic/English) electronics store with full RTL support and SAR pricing (catalogue, search with suggestions, cart, wishlist, and Stripe checkout with webhooks) using React, Node.js/Express, and MongoDB Atlas.
- Secured accounts and orders with JWT access tokens and refresh-token cookies, email verification, password reset, admin-only routes, Joi validation, and rate limiting; added coupons and order tracking.
- Built an admin dashboard for products, categories, orders, users, and coupons with sales analytics; deployed as a Vercel serverless function with optional Redis caching, Cloudinary images, and Winston logging.

Contora: Multi-Tenant Document Management System

[API](#) · [Web](#)

- Built a multi-tenant document management system with NestJS, Angular (standalone components, signals, zoneless), and MongoDB, where each organization defines its own document types in a drag-and-drop form builder and keeps records in folders with full version history (compare and restore), check-out, and trash.
- Enforced tenant isolation with per-folder and per-record access grants, expiring public share links, rotating refresh tokens, and an append-only audit log, with cross-tenant isolation tests on every endpoint (Vitest).
- Implemented Arabic-aware full-text search and content-addressed file storage (SHA-256), deployed with Docker and nginx.
- Added Azure OpenAI features: an in-app assistant that searches and proposes actions for user confirmation, and metadata extraction that fills a new record's fields from an uploaded file.

Testly: Online Exam Platform

[Live](#) · [Code](#)

- Built a role-based exam platform for students, teachers, and admins with Node.js/Express, Angular 19 (NgRx), and MongoDB Atlas, including email confirmation, password reset, and admin approval for teacher accounts.
- Implemented server-side grading that enforces time limits and never exposes answers to the client, plus per-role dashboards (attempts, results, pass rates) with Redis caching, rate limiting, and Swagger-documented APIs.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL

Backend: Node.js, NestJS, Express.js, REST APIs, GraphQL, Microservices, Event-Driven Architecture, OpenAPI/Swagger, Zod, Joi

Frontend: Angular (Signals, standalone, zoneless), NgRx, RxJS, React, Next.js (App Router), Vite, Redux Toolkit, Zustand, TanStack Query, Tailwind CSS, shadcn/ui, MUI, Angular Material, Form.io, i18next, Leaflet, Arabic RTL

Databases & Search: MongoDB (Mongoose), PostgreSQL, MySQL, SQL Server, Redis, Elasticsearch

Messaging & Workflow: NATS, ActiveMQ, Temporal, Camunda Zeebe (BPMN)

Security & Identity: Keycloak, OAuth 2.0, OpenID Connect, JWT, RBAC, Passport.js (Google sign-in)

AI: LLM integration (Vercel AI SDK, Azure OpenAI, OpenAI, Anthropic Claude, Google Gemini), AI agents & tool calling, Claude Code, MCP

Integrations: Stripe, Moyasar, Tap, PayPal, Paymob, Cloudinary, Nodemailer

Cloud & DevOps: AWS, Azure, Vercel, Docker, Kubernetes, CI/CD, Git, Bitbucket, Linux, Nginx, KrakenD, MinIO, Unleash

Observability & Testing: Grafana, OpenTelemetry, Jest, Vitest, Supertest, Playwright, Jasmine/Karma

EDUCATION & TRAINING

MEARN Stack Diploma — Information Technology Institute (ITI)

Mar 2025 – Aug 2025

Intensive full-time programme

B.Sc. in Computer Science — Zagazig University

2019 – 2023

Zagazig, Egypt

LANGUAGES

Arabic: Native · **English:** Professional working proficiency (documentation, technical writing, daily Agile collaboration)